

Locality refers to the scope of analysis, spanning from individual neurons (e.g., feature visualization) to entire model architectures (e.g., causal abstraction). Comprehensiveness varies from partial insights into specific components to holistic explanations of model behavior.

Table 1: Taxonomy of Mechanistic Interpretability Methods

Method	Causal Nature	Phase	Locality	Comprehensiveness	Key Examples
Feature Visualization	Observation	Post-hoc	Local	Partial	Zeiler & Fergus (2014) Zimmermann et al. (2021)
Exemplar methods	Observation	Post-hoc	Local	Partial	Grosse et al. (2023) Garde et al. (2023)
Probing Techniques	Observation	Post-hoc	Both	Both	McGrath et al. (2022) Gurnee et al. (2023)
Structured Probes	Observation	Post-hoc	Both	Both	Burns et al. (2023)
Logit Lens Variants	Observation	Post-hoc	Global	Partial	nostalgebraist (2020) Belrose et al. (2023)
Sparse Autoencoders	Observation	Post-hoc	Both	Comprehensive	Cunningham et al. (2024) Bricken et al. (2023)
Activation Patching	Intervention	Post-hoc	Local	Partial	Meng et al. (2022a) Wang et al. (2023)
Path Patching	Intervention	Post-hoc	Both	Both	Goldowsky-Dill et al. (2023)
Causal Abstraction	Intervention	Post-hoc	Global	Comprehensive	Geiger et al. (2023a) Geiger et al. (2023b) Wu et al. (2023a)
Hypothesis Testing	Intervention	Post-hoc	Global	Comprehensive	Chan et al. (2022) Jenner et al. (2023)
Intrinsic Methods	–	Pre/During	Global	Comprehensive	Elhage et al. (2022a) Liu et al. (2023a)

The categorization is based on the methods’ general tendencies. Some methods can offer local and global or partial and comprehensive interpretability depending on the scope of the analysis and application. Probing techniques can range from local to global and partial to comprehensive; simple linear probes might offer local insights into individual *features*, while more sophisticated structured probes can uncover global patterns. Sparse autoencoders decompose individual neuron activations (local) but aim to disentangle features across the entire model (global). Path patching extends local interventions to global model understanding by tracing information flow across layers, demonstrating how local perturbations can yield broader insights.

In practice, mechanistic interpretability research involves both method development and their application. When applying methods to understand a model, combining techniques from multiple categories is often necessary and beneficial to build a more comprehensive understanding (Section 4.4).

4.2 Observation

Mechanistic interpretability draws from observational methods that analyze the inner workings of neural networks, with many of these methods preceding the field itself. For a detailed exploration of inner interpretability methods, refer to ([Räuker et al., 2023](#)). Two prominent categories are example-based methods and feature-based methods:

- i.)* **Exemplar methods** identify real input examples that highly activate specific neurons or layers. This helps pinpoint influential data points that maximize neuron activation within the neural network ([Grosse et al., 2023](#); [Garde et al., 2023](#); [Nanfack et al., 2024](#)).
- ii.)* **Feature visualization** encompasses techniques that generate synthetic inputs to optimize neuron activation. These visualizations reveal how neurons respond to stimuli and which features are sensitive to ([Zeiler & Fergus, 2014](#); [Zimmermann et al., 2021](#)). By inspecting the synthetic inputs that drive neuron behavior, we can hypothesize about the features encoded by those neurons.

Probing for Features. Probing (Alain & Bengio, 2016; Hewitt & Manning, 2019) involves training a classifier using the activations of a model, with the classifier’s performance subsequently observed to deduce insights about the model’s behavior and internal representations. However, the probe’s performance may often reflect its own learning capacities more than the actual characteristics of the model’s representations (Belinkov, 2021). This dilemma has led researchers to investigate the ideal balance between the complexity of a probe and its capacity to accurately represent the model’s features (Cao et al., 2021; Voita & Titov, 2020).

The *linear representation* hypothesis offers a resolution to this issue. Under this hypothesis, the failure of a simple linear probe to detect certain features suggests their absence in the model’s representations. Conversely, suppose a more complex probe succeeds where a simpler one fails. In that case, it implies that the model contains features that a complex function can combine into the target feature but that the target feature itself is not explicitly represented. Thus, the hypothesis implies that using linear probes could suffice in most cases, circumventing the complexity considerations generally associated with probing (Belinkov, 2021).

McGrath et al. (2022) analyzed chess knowledge acquisition in AlphaZero, revealing the emergence of strategic concepts during training. In language models, Gurnee et al. (2023) introduced *sparse probing* to decode internal neuron activations to understand feature representation and sparsity. They show that early layers use sparse combinations of neurons to represent many features in superposition, while middle layers seem to have dedicated *monosemantic* neurons for higher-level contextual features.

Probing is limited in drawing causal or behavioral conclusions. Its primarily observational nature focuses on how information is encoded rather than how it is used (see Figure 1), necessitating careful analysis and integration with interventional techniques (Section 4.3), or alternative approaches (Elazar et al., 2021). While in explainable AI, probing has primarily analyzed high-level concepts like linguistic representations (Tenney et al., 2019; Dalvi et al., 2019), MI aims to probe towards uncovering underlying computational processes and functionality. This shift in goals towards uncovering mechanistic computation is a nuanced distinction rather than a clear-cut line between probing in MI and the broader explainability field.

Structured Probes. While focusing on bottom-up, mechanistic interpretability approaches, we can also consider integrating top-down, concept-based structured probes with mechanistic interpretability.

Structured probes aid conceptual interpretability, probing language models for complex features like truth representations. Notably, Burns et al. (2023)’s *contrast-consistent search* identifies linear projections exhibiting logical consistency in hidden states, contrasting truth values for statements and negations.

However, structured probes face significant challenges in unsupervised probing scenarios. As Farquhar et al. (2023) showed, arbitrary features, not just knowledge-related ones, can satisfy contrast consistency equally well, raising doubts about scalability. For example, the loss may capture *simulation* of knowledge from hypothesized *simulacra* within sufficiently powerful language models rather than the models’ true knowledge. Furthermore, Farquhar et al. (2023) demonstrates self-supervised probing methods (like (Burns et al., 2023)) often detect prominent but unintended distractor features in the data. The discovered features are also highly sensitive to prompt choice, and there is no principled way to select prompts that would reliably surface a model’s true knowledge.

While structured probes primarily focus on high-level conceptual representations (Zou et al., 2023), their findings could potentially inform or complement mechanistic interpretability efforts. For instance, identifying truth directions through structured probes could help guide targeted interventions or analyze the underlying circuits responsible for truthful behavior using mechanistic techniques such as activation patching or circuit tracing (Section 4.3). Conversely, mechanistic methods could provide insights into how truth representations emerge and are computed within the model, addressing some of the challenges faced by unsupervised structured probes.

Logit Lens. The *logit lens* (nostalgebraist, 2020) provides a window into the model’s predictive process by applying the final classification layer (which projects the residual stream activation into logits/vocabulary space) to intermediate activations of the residual stream, revealing how prediction confidence evolves across

computational stages. This is possible because transformers tend to build their predictions across layers iteratively (Geva et al., 2022). Extensions of this approach include the tuned lens (Belrose et al., 2023), which trains affine probes to decode hidden states into probability distributions over the vocabulary, and the Future Lens (Pal et al., 2023), which explores the extent to which individual hidden states encode information about subsequent tokens.

Researchers have also investigated techniques that bypass intermediate computations to probe representations directly. Din et al. (2023) propose using linear transformations to approximate hidden states from different layers, revealing that language models often predict final outputs in early layers. Dar et al. (2022) present a theoretical framework for interpreting transformer parameters by projecting them into the embedding space, enabling model alignment and parameter transfer across architectures.

Other techniques focus on interpreting specific model components or submodules. The DecoderLens (Langedijk et al., 2023) allows analyzing encoder-decoder transformers by cross-attending intermediate encoder representations in the decoder, shedding light on the information flow within the encoder. The Attention Lens (Sakarvadia et al., 2023b) aims to elucidate the specialized roles of attention heads by translating their outputs into vocabulary tokens via learned transformations.

Feature Disentanglement via Sparse Dictionary Learning. Recent work suggests that the essential elements in neural networks are linear combinations of neurons representing features in superposition (Elhage et al., 2022b). To disentangle these features, researchers have developed sparse autoencoders (SAEs), which decompose neural network activations into individual component features (Sharkey et al., 2022b; Cunningham et al., 2024). This process, known as sparse dictionary learning, reconstructs activation vectors as sparse linear combinations of directional vectors within the activation space (Olshausen & Field, 1997).

The theoretical foundations of SAEs are rooted in work on *disentangled* representations. Whittington et al. (2022) demonstrate that autoencoders can recover ground truth features under conditions of feature sparsity and non-negativity. Furthermore, Garfinkle & Hillar (2019) provides guarantees for the uniqueness and stability of dictionaries for sparse representation, even in the presence of noise. These theoretical underpinnings support SAEs’ ability to uncover true, disentangled features underlying the data distribution.

In practice, SAEs stand out for their simplicity and scalability (Sharkey et al., 2022b). They incorporate sparsity regularization to encourage learning sparse yet meaningful data representations, with the precise tuning of the sparsity penalty on hidden activations critical in dictating the autoencoder’s sparsity level. We provide an overview of the SAE architecture in Figure 8.

SAEs’ dictionary features exhibit higher scores on autointerpretability metrics and increased monosemanticity (Bricken et al., 2023; Cunningham et al., 2024; Sharkey et al., 2022b). They are scalable to state-of-the-art models and can detect safety-relevant features (Templeton et al., 2024), measure feature sparsity (Deng et al., 2023), and interpret reward models in reinforcement learning-based language models (Marks et al., 2023a).

Evaluating SAE quality remains challenging due to the lack of ground-truth interpretable features. Researchers have addressed this through various approaches: Karvonen et al. (2024) proposed using language models trained on chess and Othello transcripts as testbeds, providing natural collections of interpretable features. Sharkey et al. (2022b) constructed a toy model with traceable features, while Makelov et al. (2024); Makelov (2024) compared SAE results with supervised features in large language models to demonstrate their viability.

The versatility of SAEs extends to various neural network architectures. They have been successfully applied to transformer attention layers (Kissane et al., 2024) and convolutional neural networks (Gorton, 2024). Notably, Gorton (2024) applied SAEs to the early vision layers of InceptionV1, uncovering new interpretable features, including additional curve detectors not apparent from examining individual neurons (Cammarata et al., 2020).

In circuit discovery, SAEs have shown particular promise (see also Section 4.4). He et al. (2024) proposed a circuit discovery framework alternative to activation patching (discussed in Section 4.3.1), leveraging dictionary features decomposed from all modules writing to the residual stream. Similarly, O’Neill & Bui (2024) employed discrete sparse autoencoders for discovering interpretable circuits in large language models.